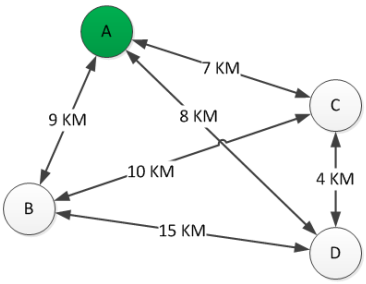
The Robert M. Buchan Department of Mining

Queen's University

MNTC 313 – Assignment 5 Version A

A typical situation in computer science and engineering is the one of computing distance matrices[1,2]. Give a total of 12 (x,y) coordinates and record the distance from each of the locations to the rest. You can use the table below as example input values, but keep in mind you will need to develop a program that can run with 12 arbitrary location coordinates.

Location	X coordinate (km)	Y coordinate (km)
1	2.1	10.0
2	5.5	8.2
3	5.5	10.1
4	10.8	3.6
5	4.0	11.3
6	2.8	7.5
7	6.3	10.0
8	6.3	1.1
9	8.0	4.0
10	4.3	7.0
11	6.3	5.1
12	10.8	3.3



Using arrays in C/C++, develop a program that satisfies the following requirements:

- Print each of the coordinate values in **two rows** separated by commas (do not include a comma after the **last value** in the **second row**).
- Calculate and output the average centre coordinate across the 12 trials (***average of X coordinates***, ***average of Y coordinates***).
- You have been given starter code that reads input coordinates into a 12 x 2 array. Create a new 12 x 12 array that stores the **distance** from each of the locations ***i*** to all the rest ***j*** (including the distance from ***i*** to ***i*** which is trivially 0), and print the values in a 12 x 12 formation with values separated by commas (do not include a comma after the last value in a row).
- Calculate and output the average distance of each row, that is the average distance from each of the locations ***i*** to all the rest ***j***. Do not take into account the trivial value 0 from ***i*** to ***i*** for this average. ***Hint: include in the sum but just divide by 11.***

[1] one of the many examples in where distance matrices are used is for solving the Traveling Salesman Problem https://en.wikipedia.org/wiki/Travelling_salesman_problem

https://tspvis.com/

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Note: A starter code file has been provided on OnQ. Download the file, and copy and paste its contents into a new .c/.cpp file in your IDE.

Reminder: Please **DO NOT** include any personal information such as name, student number or Net ID in your submission. All submissions to **Gradescope** will be run through plagiarism detection software.

With the following input into the scanf() statement:

```
2.1,10.0 5.5,8.2 5.5,10.1 10.8,3.6 4.0,11.3 2.8,7.5 6.3,10.0 6.3,1.1 8.0,4.0 4.3,7.0 6.3,5.1 10.8,3.3
```

Your output should look like:

```
(2.1,10.0),(5.5,8.2),(5.5,10.1),(10.8,3.6),(4.0,11.3),(2.8,7.5),
(6.3,10.0),(6.3,1.1),(8.0,4.0),(4.3,7.0),(6.3,5.1),(10.8,3.3)

Average center: (6.1,6.8)

Distance from point 1 to
1.- 0.0, 2.- 3.8, 3.- 3.4, 4.- 10.8, 5.- 2.3, 6.- 2.6, 7.- 4.2, 8.- 9.8, 9.- 8.4, 10.- 3.7, 11.- 6.5, 12.- 11.0
Distance from point 2 to
1.- 3.8, 2.- 0.0, 3.- 1.9, 4.- 7.0, 5.- 3.4, 6.- 2.8, 7.- 2.0, 8.- 7.1, 9.- 4.9, 10.- 1.7, 11.- 3.2, 12.- 7.2
Distance from point 3 to
1.- 3.4, 2.- 1.9, 3.- 0.0, 4.- 8.4, 5.- 1.9, 6.- 3.7, 7.- 0.8, 8.- 9.0, 9.- 6.6, 10.- 3.3, 11.- 5.1, 12.- 8.6
Distance from point 4 to
1.- 10.8, 2.- 7.0, 3.- 8.4, 4.- 0.0, 5.- 10.3, 6.- 8.9, 7.- 7.8, 8.- 5.1, 9.- 2.8, 10.- 7.3, 11.- 4.7, 12.- 0.3
Distance from point 5 to
1.- 2.3, 2.- 3.4, 3.- 1.9, 4.- 10.3, 5.- 0.0, 6.- 4.0, 7.- 2.6, 8.- 10.5, 9.- 8.3, 10.- 4.3, 11.- 6.6, 12.-
10.5
Distance from point 6 to
1.- 2.6, 2.- 2.8, 3.- 3.7, 4.- 8.9, 5.- 4.0, 6.- 0.0, 7.- 4.3, 8.- 7.3, 9.- 6.3, 10.- 1.6, 11.- 4.2, 12.- 9.0
Distance from point 7 to
1.- 4.2, 2.- 2.0, 3.- 0.8, 4.- 7.8, 5.- 2.6, 6.- 4.3, 7.- 0.0, 8.- 8.9, 9.- 6.2, 10.- 3.6, 11.- 4.9, 12.- 8.1
Distance from point 8 to
1.- 9.8, 2.- 7.1, 3.- 9.0, 4.- 5.1, 5.- 10.5, 6.- 7.3, 7.- 8.9, 8.- 0.0, 9.- 3.4, 10.- 6.2, 11.- 4.0, 12.- 5.0
Distance from point 9 to
1.- 8.4, 2.- 4.9, 3.- 6.6, 4.- 2.8, 5.- 8.3, 6.- 6.3, 7.- 6.2, 8.- 3.4, 9.- 0.0, 10.- 4.8, 11.- 2.0, 12.- 2.9
Distance from point 10 to
1.- 3.7, 2.- 1.7, 3.- 3.3, 4.- 7.3, 5.- 4.3, 6.- 1.6, 7.- 3.6, 8.- 6.2, 9.- 4.8, 10.- 0.0, 11.- 2.8, 12.- 7.5
Distance from point 11 to
1.- 6.5, 2.- 3.2, 3.- 5.1, 4.- 4.7, 5.- 6.6, 6.- 4.2, 7.- 4.9, 8.- 4.0, 9.- 2.0, 10.- 2.8, 11.- 0.0, 12.- 4.8
Distance from point 12 to
1.- 11.0, 2.- 7.2, 3.- 8.6, 4.- 0.3, 5.- 10.5, 6.- 9.0, 7.- 8.1, 8.- 5.0, 9.- 2.9, 10.- 7.5, 11.- 4.8, 12.- 0.0

Average distance from point 1: 6.1 km
Average distance from point 2: 4.1 km
Average distance from point 3: 4.8 km
Average distance from point 4: 6.7 km
Average distance from point 5: 5.9 km
Average distance from point 6: 5.0 km
Average distance from point 7: 4.9 km
Average distance from point 8: 6.9 km
Average distance from point 9: 5.1 km
Average distance from point 10: 4.3 km
Average distance from point 11: 4.4 km
Average distance from point 12: 6.8 km
```